

Past, Present and Future of “FORECASTING”

by

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Recent Trends and Buzzwords

- **Statistics** is the study of the collection, analysis, interpretation, presentation and organization of data. Statistics as a key Technology has its applications ranging from natural science, technology to social sciences and human welfare.
- **Data science** is the study of the generalizable extraction of knowledge from data, yet the key word is science.
- **Machine learning** gives computers the ability to learn without being explicitly programmed.
- **Artificial Intelligence** is the study of intelligent agents: any device that perceives its environment and takes actions that maximize its chance of success at some goal.
- **Forecasting** is estimating how the sequence of observations will continue into the future. Whether it is the rise/fall in exchange rates, the outcome of elections, or winners at the Oscars, there is sure to be something you want to know.

Reputations can be made and lost by making Forecast

“I think there is a world market for maybe five computers.”

(Chairman of IBM, 1943)

“There is no reason anyone would want a computer in their home.”

(President, DEC, 1977)

“There’s no chance that the iPhone is going to get any significant market share. No chance.”

(Steve Ballmer, CEO Microsoft, April 2007)

“We’re going to be opening relatively soon ... The virus ... will go away in April.”

(Donald Trump, February 2020)

“Prediction is very difficult, especially if it's about the future!”

- Niels Bohr, Danish Physicist & Nobel laureate in Physics.

- Forecasting has fascinated people for thousands of years, sometimes being considered a sign of divine inspiration, and sometimes being seen as a criminal activity.
- The Jewish prophet Isaiah wrote in about 700 BC "Tell us what the future holds, so we may know that you are gods." (Isaiah 41:23)
- One hundred years later, in ancient Babylon, forecasters would foretell the future based on the distribution of maggots in a rotten sheep's liver.



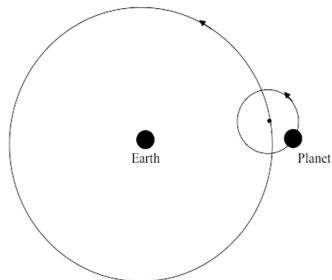
Forecasting by maggots: Clay model of sheep's liver, stored in British Museum.

- Beginning in the 800 BC, a priestess known as the Pythia would answer questions about the future at the Temple of Apollo on Greece's Mount Parnassus.
- It is said that she, the Oracle of Delphi, dispensed her wisdom in a trance – caused, some believe, by the **hallucinogenic gases** that would seep up through natural vents in the rock.



Forecasting by hallucination

- Ancients Greeks developed predictive model of the cosmos, used for astrology.
- Celestial bodies moved around the Earth in circles (the most stable and symmetric of forms).
- Motion of planets required the use of epicycles.
- Final model by Ptolemy became the official model of the church, remained unquestioned until the Renaissance (Copernicus, Galileo, Kepler).



Vagrant forecasters

Forecasters are to blame!

- Forecasters had a tougher time under the emperor Constantius, who issued a decree in AD357 forbidding anyone “to consult a soothsayer, a mathematician, or a forecaster – May curiosity to foretell the future be silenced forever.”



Codex Theodosianus 9.16.4

- A similar ban on forecasting occurred in England in 1736 when it became an offence to defraud by charging money for predictions. The British Vagrancy Act (1736) made it an offence to defraud by charging money for predictions. The punishment was three months' imprisonment with hard labour!

Forecasting. Prediction.
Divination.



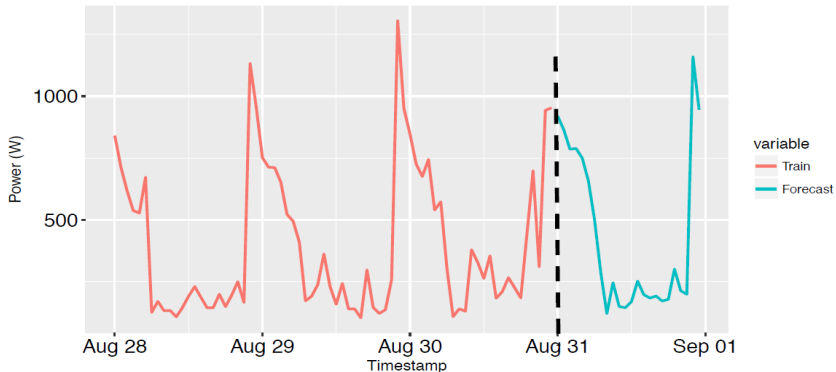
Vagrant forecasters

- **News report on 16 August 2006:** A Russian woman is suing weather forecasters for wrecking her holiday. A court in Uljanovsk heard that Alyona Gabitova had been promised 28 degrees and sunshine when she planned a camping trip to a local nature reserve, newspaper Nowyje Iswestija said.
- But it did nothing but pour with rain the whole time, leaving her with a cold. Gabitova has asked the court to order the weather service to pay the cost of her travel.

- **Time series** is a set of observations, each one being recorded at a specific time. (e.g., Forecasting of major economic variables like Annual GDP of a country, Unemployment, Inflation, Exchange rates, Production and Consumption).
- **Discrete time series** is one in which the set of time points at which observations are made is a discrete set. (e.g., All above including irregularly spaced data)
- **Continuous time series** are obtained when observations are made continuously over some time intervals. (e.g., ECG graph)
- **Forecast** is an estimate of the probability distribution of a variable to be observed in the future.

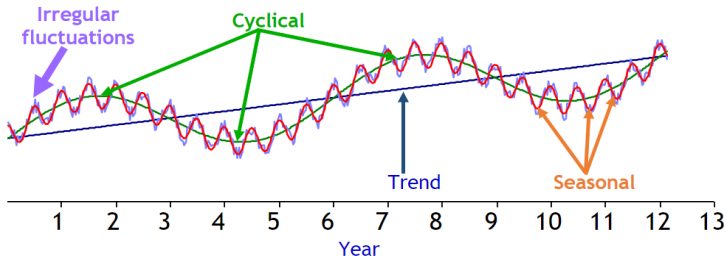
Statistical Forecasting : Assumptions

- Time series Forecasting : Data collected at regular intervals of time (e.g., Weather and Electricity Forecasting).
- Assumptions: (a) Historical Information is available; (b) Past patterns will continue in the future.
- Statistical models, though, only work in the short term, are not very good for long-term forecasting because these assumption.



Time Series Components

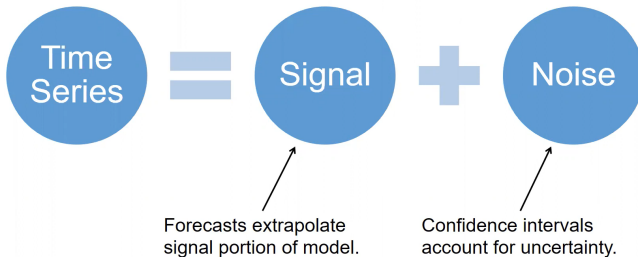
- **Trend (T_t)** : pattern exists when there is a long-term increase or decrease in the data.
- **Seasonal (S_t)** : pattern exists when a series is influenced by seasonal factors (e.g., the quarter of the year, the month, or day of the week).
- **Cyclic (C_t)** : pattern exists when data exhibit rises and falls that are not of fixed period (duration usually of at least 2 years).
- **Irregular component (I_t)**: random and unpredictable portion of the series.



Statistical Forecasting

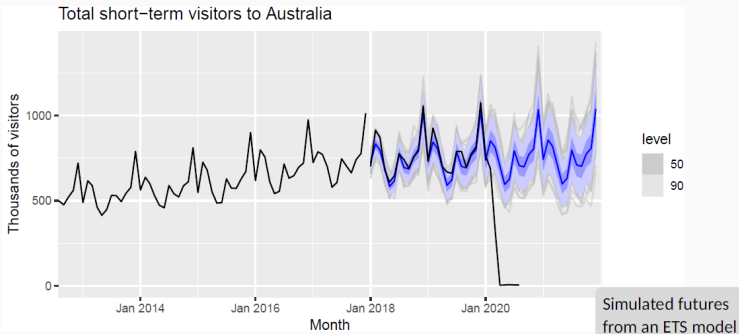
The science of forecasting got going properly in the 1980s when People realised that if you took all of the ideas that people had developed in different fields, and you thought of it as a collection of techniques and overlaid that with analytical and scientific thinking, then forecasting itself could be considered a scientific discipline.

Statistical Forecasting



Random futures

A forecast is an estimate of the probability distribution of a variable to be observed in the future.



Mathematical/Statistical models are simplifications of reality – and life is sometimes too complex to model accurately.

Exercise: Which is easiest to forecast?

- Exchange rate of USD/INR next week.
- Daily electricity demand in 3 days time.
- Time of sunrise this day next year.
- Maximum temperature tomorrow.
- Google stock price tomorrow.

Which is easiest to forecast? (Easy to Tough)

- ① Time of sunrise this day next year.
- ② Maximum temperature tomorrow.
- ③ Daily electricity demand in 3 days time.
- ④ Google stock price tomorrow.
- ⑤ Exchange rate of USD/INR next week.

How do we measure “easiest”?

What makes something easy/difficult to forecast?

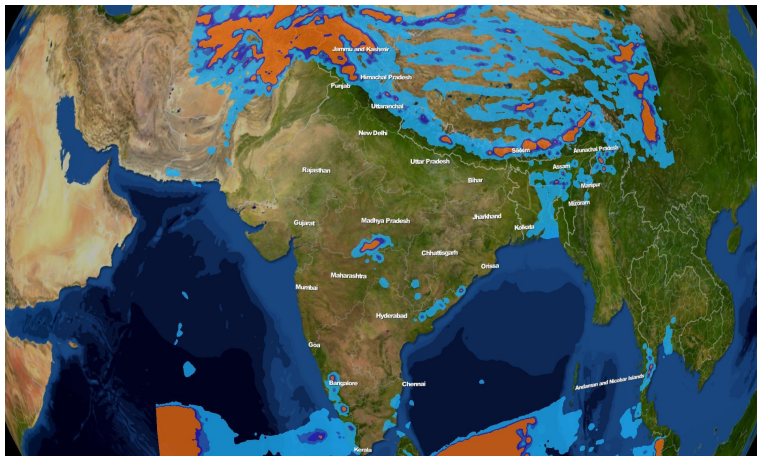
Something is **easier to forecast** if:

- We have a good understanding of the **factors** that contribute to it, and can measure them (for stock price and exchange rates causes are mostly unknown).
- There is lots of **data available**.
- The future is somewhat **similar to the past**.
- **The forecasts cannot affect the thing we are trying to forecast** (say, Warren Buffett, CEO of Berkshire Hathaway, make some comment that stock price may change!).
- **When should we give up?** When there is insufficient data? When the models give implausible forecasts?

What can we forecast? - Weather

Publicly available Weather statistics Data of Norwegian Centre for Climate Services

<https://seklima.met.no/>



What can we forecast? - Stock Price

Publicly available stock price data: <https://finance.yahoo.com/>



What can we forecast? - Sales

Time series models are extremely useful in pharmaceutical sales forecasting and in predicting demand for items.



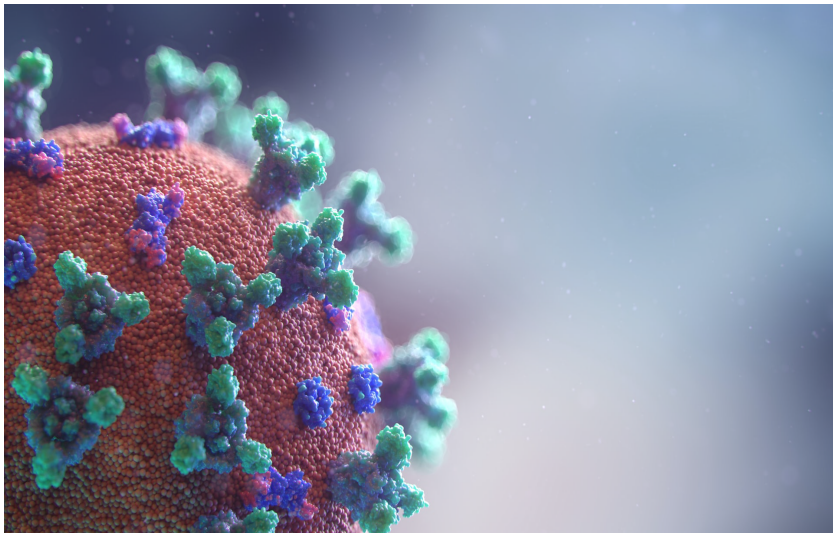
What can we forecast? - Electricity

Models for electricity demand were developed so that they could plan generation capacity.



What can we forecast? - COVID-19

Publicly available data at <https://ourworldindata.org/>



UNEMPLOYMENT RATE FORECASTING : A HYBRID APPROACH

Publication:

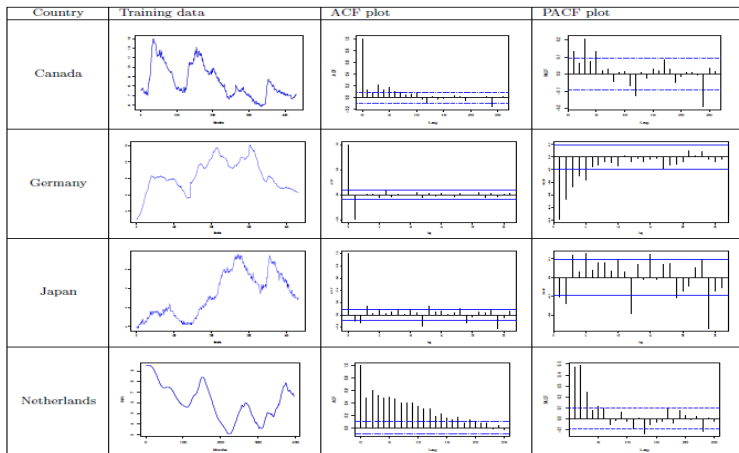
Tanujit Chakraborty, Ashis Chakraborty, Munmun Biswas, Sayak Banerjee, and Shramana Bhattacharya. "Unemployment Rate Forecasting : A Hybrid Approach". *Computational Economics* (2020) [Link](#)

Unemployment Rate Forecasting Problem

- Unemployment has always been a very focused issue causing a nation as a whole to lose its economic and financial contribution.
- Unemployment rate prediction of a country is a crucial factor for the country's economic and financial growth planning and a challenging job for policymakers.
- The unemployment rate represents the number of unemployed as a percentage of the labor force.
- Forecasting unemployment rate can be defined as the projected value for the number of unemployed people as a percentage of the labor force.
- Traditional stochastic time series models (ARIMA), as well as modern nonlinear time series techniques (Neural Net), were employed for unemployment rate forecasting previously.
- These macroeconomic data sets are mostly nonstationary and nonlinear in nature.
- Thus, it is atypical to assume that an individual time series forecasting model can generate a white noise error.

Unemployment Rate data sets

- Source : [FRED Economic Data sets](#) and [OECD data repository](#)
- Graphical analysis of training unemployment rate data sets (seasonally adjusted monthly data sets) for different countries and its corresponding ACF and PACF plots.



Previous works on Unemployment rate forecasting

- The autoregressive integrated moving average (ARIMA) is extensively utilized in constructing a forecasting model for unemployment data (Funke, *Journal of Forecasting*, 1992). [Read online](#)
- ARIMA cannot be utilized to produce an accurate model for forecasting nonlinear time series. The threshold autoregressive (TAR) model, a classical nonlinear time series model, outperformed the linear time series models for forecasting the USA unemployment rate data set (Montgomery et al., *Journal of American Statistical Association*, 1998). [Read online](#)
- ANN is found to be the most accurate in forecasting unemployment over the asymmetric business cycle for the USA, Canada, UK, France, and Japan (Pelaez, *Business Economics*, 2006). [Read online](#)
- For short term forecasting of seasonally adjusted monthly USA unemployment data sets, nonlinear models outperform the linear models. But there is an asymmetry in unemployment rate forecasting and its elimination is bound to be challenging (Nagao et al., *Finance Research Letters*, 2019). [Read online](#)
- Fractional ARIMA (FARIMA or ARFIMA) is a suitable model when long memory exists in a time series and has been applied successfully for predicting unemployment (Kartis, *Computational Economics*, 2020). [Read online](#)

AutoRegressive Integrated Moving Average (ARIMA) Model

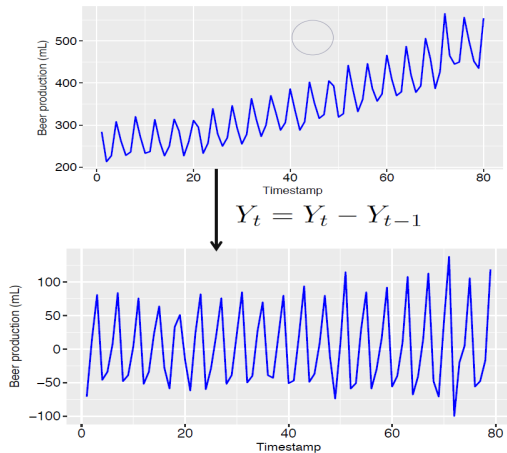
- The ARIMA model, introduced by [Box and Jenkins \(1976\)](#) ([Read online](#)), is a linear regression model indulged in tracking linear tendencies in stationary time series data.
- **AR**: autoregressive (lagged observations as inputs) **I**: integrated (differencing to make series stationary) **MA**: moving average (lagged errors as inputs).
- The model is expressed as $ARIMA(p, d, q)$ where p , d , and q are integer parameter values that decide the structure of the model.
- More precisely, p and q are the order of the AR model and the MA model respectively, and parameter d is the level of differencing applied to the data.
- The mathematical expression of the ARIMA model is as follows:

$$y_t = \theta_0 + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \varepsilon_t - \theta_1 \varepsilon_{t-1} - \theta_2 \varepsilon_{t-2} - \cdots - \theta_q \varepsilon_{t-q},$$

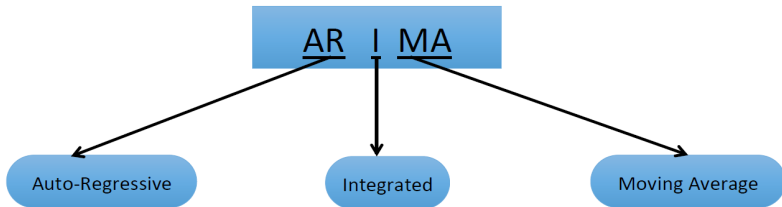
where y_t is the actual value, ε_t is the random error at time t , ϕ_i and θ_j are the coefficients of the model.

- It is assumed that ε_{t-l} ($\varepsilon_{t-l} = y_{t-l} - \hat{y}_{t-l}$) has zero mean with constant variance, and satisfies the i.i.d condition.
- Three basic Steps: Model identification, Parameter Estimation, and Diagnostic Checking.

Differencing in ARIMA Model



Differencing order
(d): Number of
times differencing is
done



$$Y_t = \beta_0 + \beta_1 Y_{t-1} + \beta_2 Y_{t-2} + \dots + \beta_p Y_{t-p} + \epsilon_t \quad [\text{Order } p]$$

$$Y_t = Y_t - Y_{t-1} \quad [\text{Order } d]$$

$$Y_t = \beta_0 + \epsilon_t + \phi_1 \epsilon_{t-1} + \phi_2 \epsilon_{t-2} + \dots + \phi_q \epsilon_{t-q} \quad [\text{Order } q]$$

ARIMA is defined by a tuple (p, d, q)

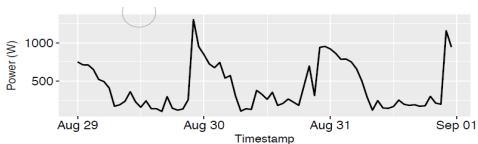
1. Auto-Correlation Function (ACF) Plot:

- Correlation coefficients of time-series at different lags
- Defines q order of MA model

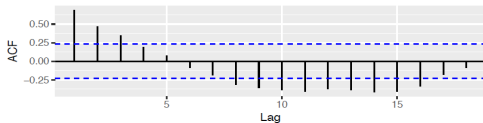
2. Partial Auto-correlation Function (PACF) Plot:

- Partial correlation coefficients of time series at different lags
- Defines p order of AR model

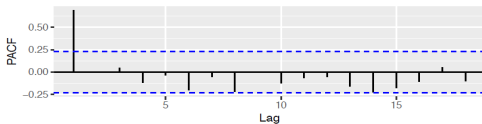
ACF / PACF Plots : Example



Data



ACF plot



PACF plot

Artificial Neural Networks (ANN) Model

- Neural nets are based on simple mathematical models of the brain, used for sophisticated nonlinear forecasting.
- ANN is composed of several perceptron-like units arranged in multiple layers (input layer, one or more hidden layer, and an output layer).
- Each hidden node computes a nonlinear transformation of its incoming inputs: Weighted linear combination followed by a nonlinear activation function.
- Learn the parameters by minimizing some loss function.
- Gradient descent backpropagation is commonly used to do this efficiently (Rumelhart, D. E., Hinton, G.E., and Williams, R. J., 1986) ([Read online](#)).

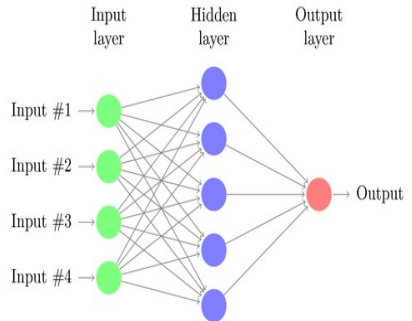


Fig: Also called a Feedforward Neural Network (FFNN) Model with one hidden layer with 5 hidden units (since there is no backward connections between layers, viz., no loops).

AutoRegressive Neural Network (ARNN) Model

- ARNN overcomes the problems of fitting ANN for time series data sets like the choice on the number of hidden neurons, and its black box nature ([Faraway and Chatfield, Journal of Royal Statistical Society - C, 1998](#)) ([Read online](#)).
- ARNN model is a nonlinear time series model which uses lagged values of the time series as inputs to the neural network.
- $ARNN(p, k)$ is a feed-forward neural network having one hidden layer with p lagged inputs and k nodes in the hidden layer.
- Thus, ARNN model with one hidden layer with the following mathematical form:

$$\hat{x}_t = \phi_0 \left\{ w_{c0} + \sum_h w_{h0} \phi_h \left(w_{ch} + \sum_i w_{ih} x_{t-j_i} \right) \right\}$$

where $\{w_{ch}\}$ denotes the the connecting weights and ϕ_i is the activation function.

- An $ARNN(p, k)$ model uses p as the optimal number of lags (calculated based on the AIC value) for an $AR(p)$ model and k is set to $k = \left\lceil \frac{(p+1)}{2} \right\rceil$ for non-seasonal data sets.

Motivation for Hybrid Techniques

- A Forecaster wants the ARIMA model error series to be composed by i.i.d. random shocks or unpredictable or unsystematic terms with zero mean and constant variance, reflecting the piece of variability for which no reduction is possible (Proietti, Computational Statistics & Data Analysis, 2002) ([Read online](#)).
- However, due to model mis-specification or to disturbances introduced in the stochastic process after forecasters elaboration, this (white noise) assumption may be violated during application phase.
- If the information underlying the error series is modeled, the performance of the original forecaster can be improved (Zhang, Neurocomputing, 2002) ([Read online](#)).
- Also, various hybrid models were developed for stock price forecasting, see (Pai and Lin, Omega, 2005) ([Read online](#)) and (Hajirahimi and Khashei, Neural Processing Letters, 2020) ([Read online](#)).
- Forecast combinations: Clemen (International journal of forecasting, 1989) stated "The results have been virtually unanimous: combining multiple forecasts leads to increased forecast accuracy... In many cases one can make dramatic performance improvements by simply averaging the forecasts" ([Read online](#)).

Proposed Hybrid Model

- An integrated approach based on linear and non-linear models is proposed that can predict unemployment more accurately.
- The hybrid forecasting method is asymptotically stationary and applied to Canada, Germany, Japan, Netherlands, New Zealand, Sweden and Switzerland unemployment data sets.
- The complexity in the time series is composed of a linear component (L) and a nonlinear component (N). We assume two models to analyse such a time series, an additive model ($L + N$).
- **Step 1:** The ARIMA model is used to model the linear component of time series.
- **Step 2:** The residuals (errors) from the ARIMA model are obtained. The residuals are modeled by the ARNN model.
- **Step 3:** the forecasts of the hybrid model are obtained by adding (multiplying) the forecasted values of linear (ARIMA) and nonlinear (ARNN) models.

Proposed Additive Hybrid Model

- The proposed hybrid ARIMA-ARNN model (Y_t) can be represented as $Y_t = L_t + N_t$, where L_t is the linear part and N_t is the nonlinear part of the hybrid model.
- Both L_t and N_t are estimated from the data set.
- Let, $\hat{L}_t$ be the forecast value of the ARIMA model at time t and ε_t represent the residual at time t as obtained from the ARIMA model, i.e., $\varepsilon_t = Z_t - \hat{Y}_t$.
- The residuals are modeled by the ARNN model and can be represented as follows $\varepsilon_t = f(\varepsilon_{t-1}, \varepsilon_{t-2}, \dots, \varepsilon_{t-n}) + \varsigma_t$, where f is a nonlinear function modeled by the ARNN approach and ς_t is the random error.
- Therefore, the combined forecast is $\hat{Y}_t = \hat{L}_t + \hat{N}_t$, where, $\hat{N}_t$ is the forecast value of the ARNN model.

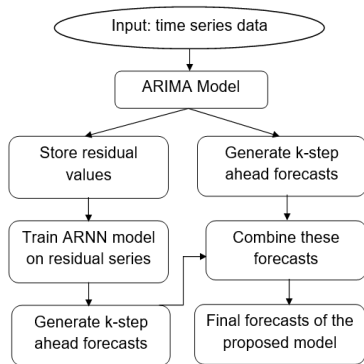


Fig: Graphical Representation of Hybrid ARIMA + ARNN Model

Asymptotic Stationarity of the Proposed Additive Model

ARIMA model has the in-built mechanism to transform a nonstationary time series into a stationary one and then it models the remainder by a stationary process. This is done by simple differencing to transform nonstationary ARIMA into stationary.

Consider the stochastic difference equation:

$$\varepsilon_t = f(\varepsilon_{t-1}, \varepsilon_{t-2}, \dots, \varepsilon_{t-p}, \theta) + \varsigma_t, \quad (0.1)$$

where ς_t is an i.i.d. white noise and $f(\cdot, \theta)$ is a feedforward neural network with weight parameter θ . This is called an ARNN process of order p and has k hidden nodes in its one hidden layer. Thus, we refer the model as ARNN(p, k) model.

We consider the following architecture:

$$f(\underline{\varepsilon}) = c_0 + \sum_{i=1}^k w_i \sigma(a_i + b'_i \underline{\varepsilon}) \quad (0.2)$$

Let ε_t denote a time series generated by a nonlinear autoregressive process as defined in (0.1). Let $E(\varepsilon_t) = 0$, then f equals to the conditional expectation $E(\varepsilon_t | \varepsilon_{t-1}, \dots, \varepsilon_{t-p})$ is the best prediction for ε_t in the L_2 -minimization sense.

We use the following notation:

$$z_{t-1} = (\varepsilon_{t-1}, \dots, \varepsilon_{t-p})'; F(z_{t-1}) = (f(z_{t-1}), \varepsilon_{t-1}, \dots, \varepsilon_{t-p+1})'; \hat{\varsigma}_t = (\varsigma_t, 0, \dots, 0)'$$

Then we can write scalar AR(p) model in (0.1) as a first-order vector model,

$$z_t = F(z_{t-1}) + \hat{\varsigma}_t \quad (0.3)$$

with $z_t, \hat{\varsigma}_t \in \mathbb{R}^p$.

Definition (Geometric ergodicity, Chan & Tong, 1985, AAP)

Let $\{z_t\}$, a markov chain, is said to be geometrically ergodic if there exists a probability measure $\Pi(A) = \lim_{t \rightarrow \infty} P(\varepsilon_t \in A)$ on the state space $(\mathbb{R}^p, \mathbb{B}, \mathbb{P})$, where $\mathbb{B}$ are Borel set on $\mathbb{R}^p$ and $\mathbb{P}$ be the Lebesgue measure, and for $\rho > 1$ and for all $z \in \mathbb{R}^p$,

$$\lim_{n \rightarrow \infty} \rho^n \|P\{z_{t+n} \in A | z_t = z\} - \Pi(A)\| = 0$$

where $\|\cdot\|$ denotes the total variation and $P\{z_{t+n} \in A | z_t = z\}$ denote the probability of going from point z to set $A \in \mathbb{B}$ in n steps.

On Asymptotic Stationarity

If the markov chain is geometrically ergodic then its distribution will converge to the stationary distribution Π and the corresponding time series will be called asymptotically stationary (Chan & Tong, 1985, [Advances in Applied Probability](#)) ([Read online](#)).

It is also important to note that all neural network activation functions (like logistics or tan-hyperbolic) are continuous and compact functions and must have a bounded range.

Lemma (Chakraborty et al., Computational Economics, 2020)

Suppose $\{z_t\}$ is defined as in (0.1) and (0.3), F be a compact set can be decomposed as $F = F_m + F_n$, and the following conditions hold:

- (i) $F_m(\cdot)$ is continuous and homogeneous and $F_n(\cdot)$ is bounded;*
- (ii) $E|\varsigma_t| < \infty$ and probability distribution function of ς_t is positive everywhere in $\mathbb{R}$;*

then $\{z_t\}$ is geometrically ergodic.

Theorem (Chakraborty et al., Computational Economics, 2020)

Let $E|\varsigma_t|^{1+\delta} < \infty$ for all $\delta > 1$ and the probability density function of ς_t is positive everywhere in $\mathbb{R}$ and $\{\varepsilon_t\}$ and $\{z_t\}$ are defined as in (0.1) and (0.3). Then if f is a nonlinear neural network as defined in (0.2), then $\{z_t\}$ is geometrically ergodic and $\{\varepsilon_t\}$ is asymptotically stationary.

Theoretical results on asymptotic stationarity is important for predictions over larger intervals of time, for example, one might train the network on an available sample and then use the trained network to generate new data with similar properties.

The asymptotic stationarity guarantees that the proposed model cannot have growing variance with time.

Performance metrics such as **mean absolute error (MAE)**, **root mean square error (RMSE)**, and **Mean Absolute Percent Error (MAPE)** are used to evaluate the performances of different forecasting models for the unemployment rate data sets:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}; \quad MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|; \quad MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|,$$

where y_i is the actual output, $\hat{y}_i$ is the predicted output, and n denotes the number of data points.

By definition, the lower the value of these performance metrics, the better is the performance of the concerned forecasting model.

Table: Performance metrics for different forecasting models on the [Canadian unemployment rate](#) (monthly) data

Model	1-Year ahead forecast			3-Year ahead forecast		
	RMSE	MAE	MAPE	RMSE	MAE	MAPE
ARIMA	0.133	0.115	1.623	0.847	0.685	9.708
ANN	0.137	0.117	1.095	0.837	0.614	9.365
ARNN	0.126	0.113	1.084	0.801	0.613	9.247
SVM	0.273	0.248	1.915	0.998	0.740	10.92
Hybrid ARIMA-SVM	0.145	0.135	1.135	0.835	0.711	9.595
Hybrid ARIMA-ANN	0.118	0.108	1.017	0.638	0.615	8.387
Proposed Hybrid ARIMA-ARNN	0.106	0.098	0.838	0.627	0.601	8.017

Table: Performance metrics for different forecasting models on the [Germany unemployment rate](#) (monthly) data

Model	1-Year ahead forecast			3-Year ahead forecast		
	RMSE	MAE	MAPE	RMSE	MAE	MAPE
ARIMA	0.098	0.092	1.490	0.361	0.303	5.177
ANN	0.127	0.120	4.295	0.564	0.505	7.394
ARNN	0.104	0.099	6.783	0.569	0.533	6.365
SVM	0.101	0.099	1.594	0.566	0.509	6.272
Hybrid ARIMA-SVM	0.090	0.089	1.537	0.360	0.305	5.120
Hybrid ARIMA-ANN	0.082	0.096	1.558	0.306	0.297	4.243
Proposed Hybrid ARIMA-ARNN	0.077	0.071	1.068	0.300	0.291	4.156

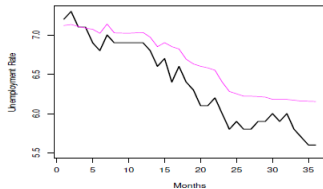
Table: Performance metrics for different forecasting models on the [Japan unemployment rate](#) (monthly) data

Model	1-Year ahead forecast			3-Year ahead forecast		
	RMSE	MAE	MAPE	RMSE	MAE	MAPE
ARIMA	0.300	0.272	6.985	0.586	0.502	14.621
ANN	0.260	0.229	6.518	0.469	0.413	11.627
ARNN	0.269	0.236	6.713	0.483	0.426	11.934
SVM	0.221	0.192	5.902	0.469	0.421	10.334
Hybrid ARIMA-SVM	0.226	0.187	5.377	0.386	0.336	9.722
Hybrid ARIMA-ANN	0.191	0.172	4.987	0.370	0.321	9.317
Proposed Hybrid ARIMA-ARNN	0.222	0.186	5.348	0.371	0.326	9.468

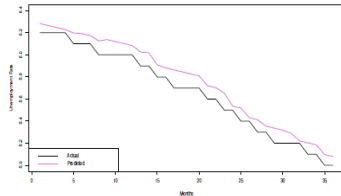
Table: Performance metrics for different forecasting models on the [Netherlands unemployment rate](#) (monthly) data

Model	1-Year ahead forecast			3-Year ahead forecast		
	RMSE	MAE	MAPE	RMSE	MAE	MAPE
ARIMA	0.144	0.119	2.003	0.306	0.270	5.671
ANN	0.228	0.174	2.906	1.304	1.048	17.935
ARNN	0.249	0.192	3.177	0.938	0.784	14.518
SVM	0.228	0.174	2.906	1.304	1.048	17.935
Hybrid ARIMA-SVM	0.145	0.120	2.023	0.308	0.272	5.706
Hybrid ARIMA-ANN	0.143	0.118	2.002	0.306	0.270	5.668
Proposed Hybrid ARIMA-ARNN	0.140	0.114	1.192	0.300	0.264	5.529

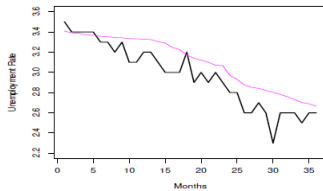
Forecast Curves



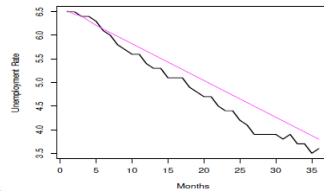
a



b



c

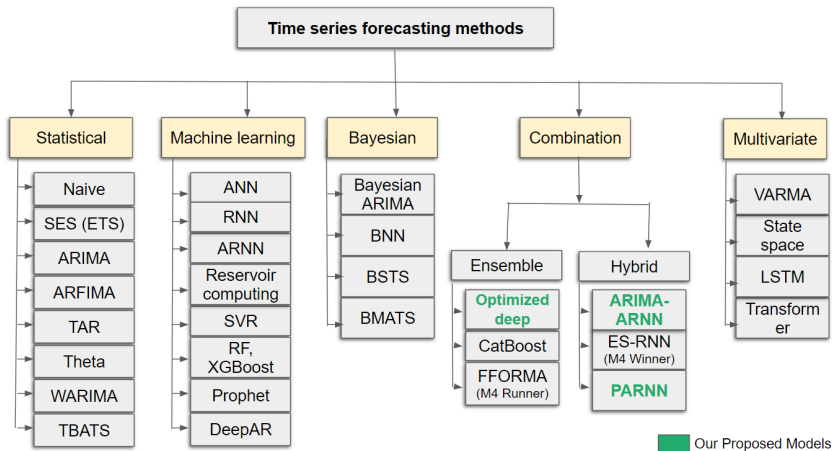


d

Fig: Actual vs Predicted (based on hybrid ARIMA-ARNN model) forecasts for the test data sets of the Canada (a), Germany (b), Japan (c), Netherlands (d)

- In practice, it is often challenging to determine whether a time series under study is generated from a linear or nonlinear underlying process.
- The additive hybridization approach studies the relationship between linear and nonlinear components of the econometric time series.
- Proposed model supposes that the residuals from the linear model will contain only the nonlinear relationship. However, one may not always guarantee that the residuals of the linear component may comprise valid nonlinear patterns.
- Also, these models supposed that the linear and nonlinear patterns of a time series can be separately modeled by different models and then the forecasts can be combined together and this may degrade performance, if it is not true.

List of Various Forecasting Models



A recently published survey paper: [Nowcasting of COVID-19 confirmed cases: Foundations, trends, and challenges](#) (Chakraborty et al., Modelling, Control and Drug Development for COVID-19 Outbreak Prevention, 2021)

- ① Based on empirical data
- ② Objective measure of uncertainty.
- ③ Computable, Replicable, Testable.
- ④ Able to compute prediction intervals.

- ① Smart algorithms, few assumptions and applied to huge data sets.
- ② Solve problems which traditional statistical methods can't handle (largely due to size of data sets).
- ③ Strong emphasis on out-of-sample predictive performance (the test data).
- ④ Recognition that many problems are about prediction not p-values.

- What about interpretable neural networks and deep learning?
- Will we ever be able to forecast “black swans”?
- Does more data mean better forecasts?
- Are there many jobs for forecasters?
- Can you design some new hybrid / ensemble / Bayesian Model Averaging-based forecasting techniques for oil price forecasting?

FINAL COMMENTS (ANONYMOUS)

“A good forecaster is not smarter than everyone else, he merely has his ignorance better organised.”

